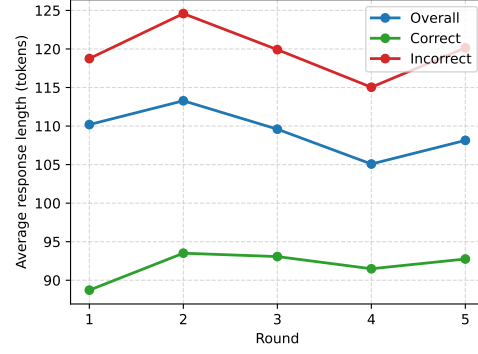
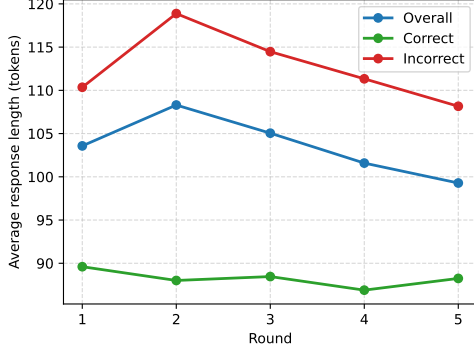




(a) Average response length, *Boosting (A)*

(b) Average response length, *Boosting (B)*

Figure 4: Average length of responses to GSM8K test set problems across rounds for *Boosting* experiments.

C Examples

In Tables 3 and 4 we present specific examples of the model improving its answer to a particular question across rounds.

<i>Question</i>	Bill starts on the 3rd floor. He rides the elevator up to the floor that's equal to 4 times his starting floor plus 6. What floor is Bill on now?
<i>Response (round 1)</i>	Let's use algebra to solve this problem. Let x be the number of floors Bill is on now. The number of floors Bill is on now is equal to $4x + 6$. We know that Bill starts on the 3rd floor, so $x = 3$. Therefore, Bill is on the 3rd floor now. The answer is 3. ✗
<i>Response (round 2)</i>	Let's use algebra to solve this problem. Let x be the floor Bill is on now. The floor Bill is on now is equal to $4x + 6$. We know that Bill starts on the 3rd floor, so $x = 3$. Therefore, Bill is on the 3rd floor now. The answer is 3. ✗
<i>Response (round 3)</i>	Let us solve it using algebra. Let x be the floor Bill is on now. The floor he rides up to is $4x + 6$. Therefore, the floor he rides up to is equal to the floor he starts on plus 6. So, we have the equation: $x = 3 + 6$ $x = 9$ Therefore, Bill is on the 9th floor now. The answer is 9. ✗
<i>Response (round 4)</i>	Let us solve it using algebra. Let x be the floor Bill is on now. The floor he is on now is $4x + 6$. We know that he starts on the 3rd floor, so $x = 3$. Therefore, the floor he is on now is $4(3) + 6 = 12 + 6 = 18$. Bill is on the 18th floor now. The answer is 18. ✓
<i>Response (round 5)</i>	Let us solve it using algebra. Let x be the floor Bill is on now. The floor he is on now is $4x + 6$. We know that he starts on the 3rd floor, so $x = 3$. Therefore, the floor he is on now is $4(3) + 6 = 12 + 6 = 18$. Bill is on the 18th floor now. The answer is 18. ✓

Table 3: Tracking responses to problem id=253 from the GSM8K test set, across rounds of a *Boosting (A)* run. The model always sets up the right expression ($4x + 6$), but in the first two rounds reports $x = 3$ as the answer instead of substituting it into the expression. The correct substitution is done in round 4, and the correct answer is maintained in round 5.

<i>Question</i>	The Kennel house keeps 3 German Shepherds and 2 Bulldogs. If a German Shepherd consumes 5 kilograms of dog food and a bulldog consumes 3 kilograms of dog food per day. How many kilograms of dog food will they need in a week?
<i>Response (round 1)</i>	Let's solve this problem using algebra. **Step 1: Find the total amount of dog food consumed by the German Shepherds in a week:** * German Shepherds: 3 dogs * Dog food consumed per day: 5 kilograms * Dog food consumed in a week: 3 dogs * 5 kilograms/dog = 15 kilograms **Step 2: Find the total amount of dog food consumed by the Bulldogs in a week:** * Bulldogs: 2 dogs * Dog food consumed per day: 3 kilograms * Dog food consumed in a week: 2 dogs * 3 kilograms/dog = 6 kilograms **Step 3: Find the total amount of dog food consumed in a week:** * Total dog food consumed: 15 kilograms + 6 kilograms = 21 kilograms Therefore, the kennel house will need 21 kilograms of dog food in a week. The answer is 21. ✗
<i>Response (round 2)</i>	Let's calculate the total amount of dog food needed in a week. **German Shepherds:** - Number of German Shepherds: 3 - Dog food consumption per day: 5 kg - Dog food consumption in a week: 3 * 7 = 21 kg **Bulldogs:** - Number of Bulldogs: 2 - Dog food consumption per day: 3 kg - Dog food consumption in a week: 2 * 7 = 14 kg **Total dog food consumption in a week:** - Total dog food consumption in a week: 21 kg + 14 kg = 35 kg Therefore, the kennel house will need 35 kilograms of dog food in a week. The answer is 35. ✗
<i>Response (round 3)</i>	Let's solve this problem using algebra. **German Shepherds:** * Number of German Shepherds: 3 * Dog food consumed per day: 5 kilograms * Dog food consumed in a week: 3 * 7 = 21 kilograms **Bulldogs:** * Number of Bulldogs: 2 * Dog food consumed per day: 3 kilograms * Dog food consumed in a week: 2 * 7 = 14 kilograms **Total dog food consumed in a week:** * Total dog food consumed in a week: 21 kilograms + 14 kilograms = 35 kilograms Therefore, the Kennel house will need 35 kilograms of dog food in a week. The answer is 35. ✗
<i>Response (round 4)</i>	Let's calculate the total dog food needed per day: * German Shepherds: 3 * 5 = 15 kilograms * Bulldogs: 2 * 3 = 6 kilograms * Total dog food needed per day: 15 + 6 = 21 kilograms Now, let's calculate the total dog food needed in a week: * 7 days * 21 kilograms/day = 147 kilograms Therefore, the kennel house will need 147 kilograms of dog food in a week. The answer is 149. ✗
<i>Response (round 5)</i>	Let's calculate the total dog food consumption per day: * German Shepherds: 3 * 5 = 15 kilograms * Bulldogs: 2 * 3 = 6 kilograms * Total: 15 + 6 = 21 kilograms Now, let's calculate the total dog food consumption in a week: * 7 days * 21 kilograms/day = 147 kilograms Therefore, the Kennel house will need 147 kilograms of dog food in a week. The answer is 147. ✓

Table 4: Tracking responses to problem id=1266 from the GSM8K test set, across rounds of a *Boosting (B)* run. The round 1 response gets the correct amount per day (21), but does not multiply to get the correct value for the week. Rounds 2 and 3 fail to incorporate the amount of food eaten per dog. Round 4 gets the correct approach, but fails the final multiplication step, which is corrected in the final round.

D Prompts

D.1 GSM8K

As an expert problem solver solve step by step the following mathematical questions.

Q:

Tina makes \$18.00 an hour. If she works more than 8 hours per shift, she is eligible for overtime, which is paid by your hourly wage + $\frac{1}{2}$ your hourly wage. If she works 10 hours every day for 5 days, how much money does she make?

A:

Here's how to calculate Tina's earnings:

****Regular Time:****

- Hours per shift: 8 hours
- Wage per hour: \$18.00
- Regular pay per shift: 8 hours * \$18.00/hour = \$144.00

****Overtime:****

- Overtime hours per shift: 10 hours - 8 hours = 2 hours
- Overtime pay per hour: $\$18.00 + (\$18.00 / 2) = \$27.00$
- Overtime pay per shift: 2 hours * \$27.00/hour = \$54.00

****Total per day:****

- Regular pay + overtime pay: $\$144.00/\text{shift} + \$54.00/\text{shift} = \$198.00/\text{day}$

****Total for 5 days:****

- 5 days * \$198.00/day = \$990.00

****Therefore, Tina will make \$990.00 in 5 days.****

The answer is 990.

Q:

Abigail is trying a new recipe for a cold drink. It uses $\frac{1}{4}$ of a cup of iced tea and 1 and $\frac{1}{4}$ of a cup of lemonade to make one drink. If she fills a pitcher with 18 total cups of this drink, how many cups of lemonade are in the pitcher?

A:

Ambiguity in the Problem Statement: